Notes 2

1. What is an Operating System?

An operating system is the main software that controls the computer's hardware and allows users to run programs. The OS relies on the kernel, which is the core component that manages communication between hardware and software.

2. What is a kernel?

The kernel is the core part of the operating system that controls the computer at a low level. The kernel manages things like the CPU, memory, and hardware, and make sure programs can run and communicate with each other properly.

3. Which other parts aside from the kernel identify an OS?

Other parts of an operating system besides the kernel include the Command-line shells, which allows users to interact with the system using tet commands; Graphical user interfaces (GUIs), which provide visual interaction through icons and menus; Utility and productivity programs, which help users perform tasks like editing files or browsing the web; and Libraries which provide shared code that applications rely on.

4. What is linux and linux distribution?

Linux is a Unix-like operating system built around the Linux kernel that provides a complete environment for running applications, managing hardware, and performing system tasks on a computer.

A linux distribution is a complete operating system built using the linux kernel. Linux distribution includes the kernel along with core system tools, libraries, a graphical interface, system utilities, and additional software needed to run the computer.

Linux distribution is made up of several components including: The Linux kernel, which is the core of the system; Core Unix tools, such as the GNU toolset; a graphical system, like the X Window System and a desktop environment; system and distribution-specific tools, which manage startup and system behavior; an installer and package manager, which handle software installation and updates; and then additional software such as server or user applications. Different distribution may use different versions but they all rely on the Linux kernel.

5. List at least 4 linux characteristics:

Linux is an open-source, allowing users to view and modify the source code. Linux is free to use with no licensing cost ; it is Unix-like that follows the Unix-design principles and tools; and finally its highly customizable, giving users control over system behavior.

6. What is Debian?

Debian is a linux operating system created by an all-volunteer organization dedicated to developing free software and supporting the free software community. Started in 1993, Debian is one of the oldest and most influential Linux distributions and has served as the foundation for many other operating systems due to its stability and reliability.

7. List and define the different types of licensing agreements

The main types of software licensing agreements are open-source, closed-source, and free software. Open-source software includes source code and can be modified and redistributed. Closed-source software does not provide access to the source code, and users are restricted from modifying it. Freeware and shareware are examples of closed-source software. Free software includes source code and allows users the freedom to run, modify, and share the software.

8. What is Free Software? Define the 4 freedoms.

Free software is a software that gives users control over how the software is used, studied, modified, and shared. Free software is defined by the Free Software Foundation which was founded in 1985 to promotes software freedom rather than profit or restriction. Free Software is often distributed under licenses like the GNU General Public License, which ensures these freedoms are protected.

The four freedoms are Freedom 0, the freedom to use the software for any purpose; Freedom 1, the freedom to study the source code and modify it as needed; Freedom 2, the freedom to redistribute copies of the software; and Freedom 3, the freedom to distribute modified versions of the software.

9. What is virtualization?

Virtualizations is a technology that creates virtual versions of computers using one physical machine. It allows multiple operating systems to run at the same time without dual booting. This reduces hardware costs and makes testing, backups and system recovery easier.